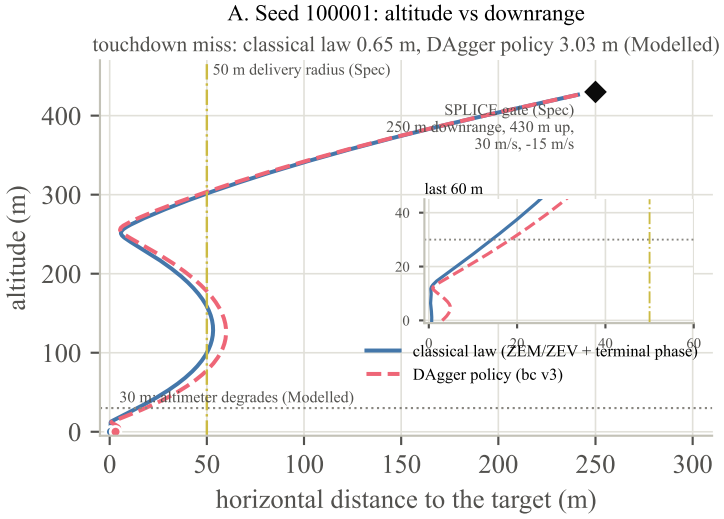
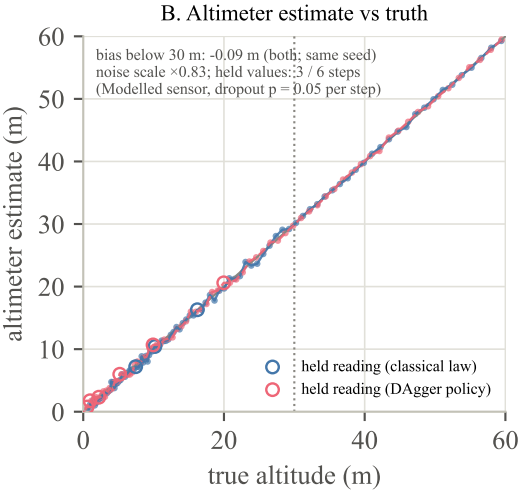
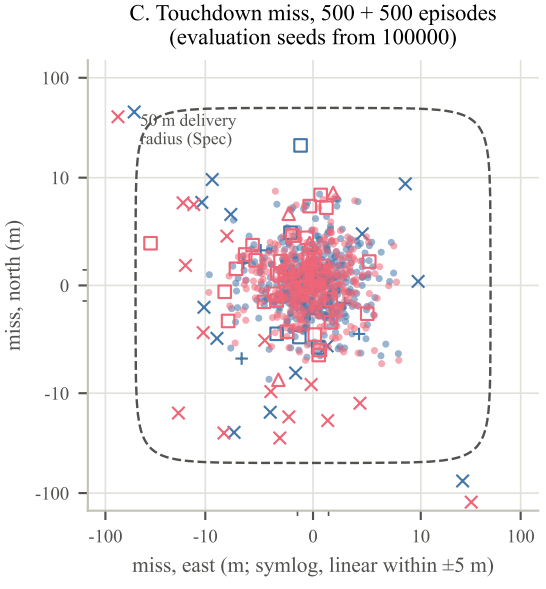
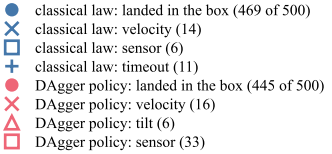
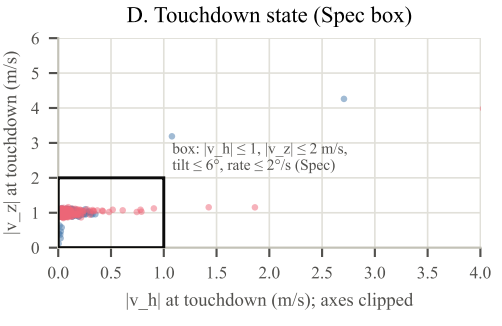

A. Seed 100001: altitude vs downrange
touchdown miss: classical law 0.65 m, DAgger policy 3.03 m (Modelled)
50 m delivery radius (Spec)
SPLICE gate (Spec)
250 m downrange, 430 m up,
30 m/s, -15 m/s
last 60 m
classical law (ZEM/ZEV + terminal phase)
DAgger policy (bc v3)
30 m altimeter degrades (Modelled)
altitude (m)
horizontal distance to the target (m)


B. Altimeter estimate vs truth
bias below 30 m: -0.09 m (both; same seed)
noise scale ×0.83; held values: 3 / 6 steps
(Modelled sensor, dropout p = 0.05 per step)
held reading (classical law)
held reading (DAgger policy)
altimeter estimate (m)
true altitude (m)


C. Touchdown miss, 500 + 500 episodes
(evaluation seeds from 100000)
50 m delivery
radius (Spec)
miss, north (m)
miss, east (m; symlog, linear within ±5 m)


classical law: landed in the box (469 of 500)
classical law: velocity (14)
classical law: sensor (6)
classical law: timeout (11)
DAgger policy: landed in the box (445 of 500)
DAgger policy: velocity (16)
DAgger policy: tilt (6)
DAgger policy: sensor (33)


D. Touchdown state (Spec box)
box: |v_h| ≤ 1, |v_z| ≤ 2 m/s,
tilt ≤ 6°, rate ≤ 2°/s (Spec)
|v_z| at touchdown (m/s)
|v_h| at touchdown (m/s); axes clipped